DIFFERENTIATION FORMULAS

If u and v are the functions of x while a, c, n denote the constants and we assume that derivatives of u and v with respect to x exist, then.

1. $\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}$

2. $\frac{d}{dx}(cv) = c\frac{dv}{dx}$

3. $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$

4. $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v(du/dx) - u(dv/dx)}{v^2}$

5. $\frac{d}{dx}(x^n) = nx^{n-1}$

6. $\frac{d}{dx}(u^n) = nu^{n-1} \cdot \frac{du}{dx}$

7. $\frac{d}{dx}(a^x) = a^x \ln a$

8. $\frac{d}{dx}(e^x) = e^x$

9. $\frac{d}{dx}(\ln x) = \frac{1}{x}$

10. $\frac{d}{dx}(\sin x) = \cos x$

11. $\frac{d}{dx}(\cos x) = -\sin x$

12. $\frac{d}{dx}(\tan x) = \sec^2 x$

13. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

14. $\frac{d}{dx}(\sec x) = \sec x \tan x$

15. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

16. $\frac{d}{dx}\left(\sin^{-1}\frac{x}{a}\right) = \frac{1}{\sqrt{a^2 - x^2}}$

17. $\frac{d}{dx}\left(\cos^{-1}\frac{x}{a}\right) = \frac{-1}{\sqrt{a^2 - x^2}}$

18. $\frac{d}{dx}\left(\tan^{-1}\frac{x}{a}\right) = \frac{a}{a^2 + x^2}$

19. $\frac{d}{dx}\left(\cot^{-1}\frac{x}{a}\right) = \frac{-a}{a^2 + x^2}$

20. $\frac{d}{dx}\left(\sec^{-1}\frac{x}{a}\right) = \frac{a}{x\sqrt{x^2 - a^2}}$

21. $\frac{d}{dx}\left(\operatorname{cosec}^{-1}\frac{x}{a}\right) = \frac{a}{x\sqrt{x^2 - a}}$

INTEGRATION FORMULA

If u and v are functions of x and a, b, c, n constants we have the following integration formulas where the constant of integration has been omitted.

1. $\int (u \pm v)\,dx = \int u\,dx \pm \int v\,dx$
2. $\int cu\,dx = c\int u\,dx$
3. $\int u\,dv = uv - \int v\,du$ (integration by parts)
4. $\int x^n\,dx = \frac{x^{n+1}}{n+1}$, where $n \neq -1$
5. $\int x^{-1}\,dx = \int \frac{dx}{x} = \ln x$
6. $\int a^{bx}\,dx = \frac{a^{bx}}{b \ln a}$, $a \neq 0, 1$
7. $\int e^{ax}\,dx = \frac{1}{a}e^{ax}$
8. $\int \sin x\,dx = -\cos x$
9. $\int \cos x\,dx = \sin x$
10. $\int \tan x\,dx = -\ln \cos x = \ln |\sec x|$
11. $\int \cot x\,dx = \ln |\sin x|$
12. $\int \sec x\,dx = \ln |\sec x + \tan x|$
13. $\int \operatorname{cosec} x\,dx = \ln |(\operatorname{cosec} x - \cot x)|$
14. $\int \sec^2 x\,dx = \tan x$
15. $\int \operatorname{cosec}^2 x\,dx = -\cot x$
16. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a}$
17. $\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$
18. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \sec^{-1}\frac{x}{a}$
19. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln\left(x + \sqrt{x^2 - a^2}\right)$
20. $\int \frac{1}{\sqrt{x + a^2}} = \ln\left(x + \sqrt{x^2 + a^2}\right)$
21. $\int e^{ax} \sin bx\,dx = \frac{1}{a^2 + b^2} e^{ax}(a \sin bx - b \cos bx$
22. $\int e^{ax} \cos bx\,dx = \frac{1}{a^2 + b^2} e^{ax}(a \cos bx + b \sin bx$
23. $\int \sinh x\,dx = \cosh x$
24. $\int \cosh x\,dx = \sinh x$

Some Standard Substitutions in Integration

1. $(ax + b)^n$; let $u = ax + b$
2. $\sqrt{ax - x^2}$ or $\left(\sqrt{a^2 - x^2}\right)^n$ or $\frac{1}{\sqrt{a^2 - x^2}}$ let $x = a \sin\theta$
3. $\frac{1}{a^2 + x^2}$ or $\frac{1}{\sqrt{x^2 + a^2}}$ let $x = a \tan$
4. $\frac{1}{\sqrt{x^2 - a^2}}$ or $\frac{1}{x\sqrt{x^2 - a^2}}$ let $x =$

DIFFERENTIAL EQUATIONS

Definition: A differential equation is an equation involving an unknown function and its derivatives.

There are two types of differential equations;

1.) Ordinary differential equation
2.) Partial differential equation.

Ordinary differential Equation: It is an equation involving one dependent variable and a derivative with respect to one independent variable.

Partial differential equation: It is an equation involving one dependent variable and a derivative with respect to at least two independent variables **OR**

A partial differential equation is an equation containing one dependent, two or more independent variables and one or more partial derivatives of the dependent variable with respect to the independent variables.

The primary focus of this course will be Ordinary differential equations.

Examples of Ordinary differential equation; (ODE)

1.) $\frac{dy}{dx} = 5x + 3$ 2.) $e^y \frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^2 = 1$

3.) $y' + 2y = e^x$ 4) $y'' + 2y' + 2y = 3\sin x$

5.) $4y'' + y''\sin x + 5xy = 0$ 6) $\left(\frac{d^2y}{dx^2}\right)^3 + 3y\left(\frac{dy}{dx}\right)^4 + y^3\left(\frac{dy}{dx}\right)^2 = 3x$

NOTATIONS AND DEFINITIONS

1.) The expressions $y', y'', y''', y^{iv}, \ldots, y^n$ are often used to represent, respectively, the first, second, third, fourth, ... nth derivatives of y with respect to the independent variable under consideration. Thus $y' = \frac{dy}{dx}$, $y'' = \frac{d^2y}{dx^2}$, $y''' = \frac{d^3y}{dx^3}$, $y^{iv} = \frac{d^4y}{dx^4}$, $\ldots$, $\frac{d^ny}{dx^n} = y^n$ if the independent

general form of an nth order ordinary differential equation is $F(x, y, y', y'', y''', \ldots, y^{(n)}) = 0$. Or $\frac{d^n y}{dx^n} = f(x, y, y', y'', \ldots y^{(n-1)})$

Variable is x. (The notation $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$ is known as Leibniz notation while y', y'', etc is called the prime notation, named after Newton)

2.) The ORDER of an ODE is the order of the highest derivative in the equation

3.) The DEGREE of the an ODE is the power of the highest order derivative in the ODE

4.) A Solution of a differential equation is the unknown function y and the independent variable x on the interval I, is a function $y(x)$ that satisfies the differential equation identically for all x in I

5.) An ordinary differential equation of order n is said to be linear if it is linear in the dependent variable y and its derivatives $y', y'', y''', y^{iv} \ldots y^{(n)}$ i.e if an equation is linear in y and also linear in the derivatives we say that the ordinary differential equation is linear, otherwise its non-linear

6) In general n^{th} order linear ordinary differential equation is of the form

$$a_0 y^n + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = f(x)$$

where $a_0, a_1, a_2, \ldots a_n$ are given functions of x or constants and $f(x)$ is a function of x.

The nth order linear ordinary differential equation is said to have constant coefficient when $a_0, a_1, a_2, \ldots, a_n$ are all constants it is said to have variable coefficient when some or all of $a_0, a_1, a_2, \ldots, a_n$ are variables ie (functions of x)

7) An ODE is said to be homogeneous if $f(x) = 0$ and to be inhomogeneous or non-homogeneous if $f(x)$ is not equal zero ie $f(x) \neq 0$

Formation of Ordinary Differential Equations

If the unknown function y is expressed as a function which contain one arbitary constant the elimination of the constant always leads to the first order ODE. (8) A differential equation in which the dependent variable does not appear explicitly is said to be autonomous otherwise it is non-autonomous. Thus in general an autonomous 1st order ODE is of the form $F(y, y') = 0$

Example: 1) Let $y = x^2 + A$, where A is a constant

$\therefore y' = 2x$

Then the function you differentiated $y = x^2 + A$ is called the complete primitive or the general solution of the first order ODE $y' = 2x \iff \frac{dy}{dx} = 2x$

2) Let $y = Ae^{-x}$ where A is a constant then

$y' = -Ae^{-x}$

observe that $y' = -y = -Ae^{-x}$

Therefore to eliminate the constant

$y' + y = 0 \Rightarrow -Ae^{-x} + Ae^{-x} = 0$

The function $y = Ae^{-x}$ is the complete premitive or the general solution of the first order ODE $y' + y = 0$

Next if y is expressed as a function which contains two (2) arbitrary constants the elimination of the constant leads to a second order ODE

Ex 3: let $y = A\cos x + B\sin x$, where A, B are constants

$y' = -A\sin x + B\cos x$

$y'' = -A\cos x - B\sin x$

observe that $y'' = -y$

$\therefore y'' + y = 0$ the function $y = A\cos x + B\sin x$ is the complete premitive or the general ~~order~~ solution of the second order ODE $y'' + y = 0$

Ex 4: let $y(x) = A\sin 2x + B\cos 2x$ where A, B are arbitrary constants, a solution of $y'' + 4y = 0$?

Differentiating y, we find

$y' = 2A\cos 2x - 2B\sin 2x$

$y'' = -4A\sin 2x - 4B\cos 2x$

observe that $y'' = -4y$

$\therefore y'' + 4y = 0$

Thus $y = A\sin 2x + B\cos 2x$ satisfies the differential equation for all values of x and is a solution on the interval $(-\infty, \infty)$.

Definition: A solution of an ODE is thus a function which when substituted in the ODE satisfies it

this solution is often refered to as the integral of the ODE.

General if y is expressed as a function which contains n arbitrary constants then n differentiations are required to eliminate the constant and reduce the relation between x and y to an nth order ODE it has to be therefore noted that the general solution of an nth order ODE must contain n arbitrary constants.

Example 5: Find the ODE which have the solution

$y = Ae^{-x} + \frac{1}{2}e^{x}$ (A = constant)

soln

$y' = -Ae^{-x} + \frac{1}{2}e^{x}$

$y + y' = e^{x}$

Ex 6: Find the ODE which have the solution

$y(x) = Ae^{-x} + Be^{2x}$

soln

$y' = -Ae^{-x} + 2Be^{2x}$

$y'' = Ae^{-x} + 4Be^{2x}$

$y' + y = 3Be^{2x}$

$y'' - y = 3Be^{2x}$

$\therefore y'' - y = y' + y$

$\Rightarrow y'' - y' - 2y = 0$

Exercise:

1. Eliminate the constants from each equation, Hence finding the ODE's that have the following solution

i $y = x + \frac{A}{x}$ (ii) $y = Ae^{2x}$ (iii) $y = Ax + \ln x$

iv $x^2 + y = Ay$ (v) $y = \sin x + \frac{A}{\sin x}$

2. If $y = Ax^2 + Bx$, show that $x^2y'' - 2xy' + 2y = 0$

3. Show that $y = Ax^2 + B$ satisfies the equation $xy'' - y' = 0$

4 If $y^2 = k\left(\frac{x-1}{x+1}\right)$, where k = constant prove that $(x^2-1)y' =$

NOTE: In many physical problem the solution of an ODE have to satisfy certain specified conditions. These conditions known as the initial boundary conditions. determines the values of the arbitrary constants in the general solution.

The term boundary condition is used when an ODE has to be solved within a specified interval say $a < x < b$ ie $x \in (a, b)$.

If time t is an independent variable of an ODE condition is prevail at a specified time say $(t=0)$ are known as initial condition.

A differential equation is called

1. Initial value problem (IVP) if the initial condition is prescribed on it.
2. Boundary value problem (BVP) if boundary condition are prescribed on it.
3. mixed initial value problem: if both condition are prescribed on it

Example, from Ex 6 above if $y(x) = Ae^{-x} + Be^{2x}$ the general solution of the ODE $y'' - y' - 2y = 0$ satisfies $y = 0$ when $x = 0$ (ie $y(0) = 0$) and $y' = -3/2$ when $x = 0$ (ie $y'(0) = -3/2$).

Then we obtain

$(x, y) = (0, 0) \quad \therefore y(0)$

$\Rightarrow \quad 0 = Ae^{-0} + Be^{2(0)}$

$\Rightarrow \quad A + B = 0 \quad - - - \quad (1)$

$y' = -Ae^{-x} + 2Be^{2x}$

and $y'(0) = -3/2$

$\Rightarrow \quad -3/2 = -Ae^{-0} + 2Be^{2(0)}$

$\Rightarrow \quad -A + 2B = -3/2 \quad - - - \quad (2)$

Solve (1) and (2) to get $A = 1/2$, $B = -1/2$

Thus, $y(x) = \frac{1}{2}e^{-x} - \frac{1}{2}e^{2x}$ is the solution of the IVP $y'' - y' - 2y = 0$, for $y'(0) = -3/2$, $y(0) = 0$

THE FIRST ORDER ODE

This equation have the general form $\frac{dy}{dx} = f(x, y)$.
We have five standard types to consider, namely

1. Separable equations (separable variables)
2. Homogeneous equations
3. Exact equations
4. Linear equations
5. Bernoulli equations

1. SEPARABLE EQUATIONS: Variables are said to be separable if the ODE has either of its standard forms

i. $f(x)\,dx + g(y)\,dy = 0$

ii. $f(x)\,dy + g(y)\,dx = 0$

from (i): $f(x)\,dx = -g(y)\,dy$

and from (ii): $\frac{dx}{f(x)} = -\frac{dy}{g(y)}$

and, in either of these later forms the variables are said to have been separated, and direct integration gives y implicitly in terms of x.

solve

Ex1. $x\,dx - y^2\,dy = 0$.

soln

$$x\,dx - y^2\,dy = 0$$

$$x\,dx = y^2\,dy$$

$$\int x\,dx = \int y^2\,dy$$

$$\frac{x^2}{2} + A = \frac{y^3}{3}$$

solving for y explicitly, we obtain the solution as

$$y^3 = \frac{3}{2}x^2 + 3A$$

$$y(x) = \left(\frac{3}{2}x^2 + B\right)^{1/3}; \quad B = 3A$$

Ex2 Solve $y' = y^2x^3$

Soln

$y' = y^2x^3$

$\frac{dy}{dx} = y^2x^3$

$\frac{dy}{y^2} = x^3dx$

$\int y^{-2}dy = \int x^3dx$

$-y^{-1} = \frac{x^4}{4} + A$

$\frac{-1}{y} = \frac{x^4}{4} + A$

$-y = \frac{4}{x^4 + 4A}$

$y(x) = \frac{-4}{x^4 + B}$, $B = 4A$

Ex3. $y' = 5y$ Solve

Soln

$\frac{dy}{dx} = 5y$

$\frac{dy}{y} = 5dx$

$\int \frac{1}{y}dy = \int 5dx$

$\ln y = 5x + A$

$e^{\ln y} = e^{5x+A}$

$y(x) = e^{5x} \cdot e^A$

$y(x) = Be^{5x}$, $B = e^A$

Ex4 Solve $\frac{dy}{dx} = \frac{y^2-1}{x}$

Soln

$\int \frac{dy}{y^2-1} = \int \frac{dx}{x}$

$\int \frac{dy}{y^2-1} = \ln x + A$

$\int \left(\frac{1}{2(y-1)} - \frac{1}{2}\left(\frac{1}{y+1}\right)\right)dy = \ln x + A$

$\frac{1}{2}[\ln(y-1) - \ln(y+1)] = \ln x + A$

$\frac{1}{2}\left[\ln\left(\frac{y-1}{y+1}\right)\right] = \ln x + A$

$\ln\sqrt{\frac{y-1}{y+1}} = \ln x + A$

$A = \ln\sqrt{\frac{y-1}{y+1}} - \ln x$

$A = \ln\left(\frac{1}{x}\sqrt{\frac{y-1}{y+1}}\right)$

$e^A = \frac{1}{x}\sqrt{\frac{y-1}{y+1}}$

$B = \frac{1}{x}\sqrt{\frac{y-1}{y+1}}$ if $B = e^A$

$Bx = \sqrt{\frac{y-1}{y+1}}$

$B^2x^2 = \frac{y-1}{y+1}$

$Cx^2 = \frac{y-1}{y+1}$ if $C = B^2$

EXERCISE: Solve by Separable variable method:

1) $y' = \frac{y+1}{x+2}$ (2) $xdx = (x^2-1)dy$

3. $e^{x+y}dy = dx$ (4) $(x+1)y' = xy$

5) $dx = x\tan y\,dy$ (6) $(x^2-1)dy = xdx$

7) $(x^2-1)dx = xy(y-1)dy$

8) $(\sin x - \cos x)dx + (\sin x + \cos x)dy = 0$

9. $2xy' = 1 - 2y + y^2$

HOMOGENOUS FIRST ORDER EQUATION

This first order ODE is said to be homogenous if it can be written in this form $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$

The function of x and y can be written in the form $f(y/x)$ if the substitution $y = vx$ is the function of x and y reduces to the function of v.

Thus let $y = vx$, where v itself is a function of x then we obtain

$$\frac{d(vx)}{dx} = f\left(\frac{vx}{x}\right)$$

$$v + x\frac{dv}{dx} = f(v)$$

$$x\frac{dv}{dx} = f(v) - v$$

or $\frac{dv}{f(v)-v} = \frac{dx}{x}$ $\Rightarrow$ Separable variable

Thus after substitution $y = vx$ in the equation we solve the equation by separating the variables.

NOTE: In the general framework of differential equation the word "homogenous" has an entirely different meaning. Only in the content of first-order differential equations does "homogenous" have the meaning defined above.

Ex1 Solve $y' = \frac{y+x}{x}$

Soln

$$\frac{dy}{dx} = \frac{y+x}{x}$$

$$\frac{d(vx)}{dx} = \frac{vx+x}{x}$$

$$v + x\frac{dv}{dx} = \frac{x(v+1)}{x}$$

$$v + x\frac{dv}{dx} = v+1$$

$$x\frac{dv}{dx} = v+1-v$$

$$x\frac{dv}{dx} = 1$$

$$\int dv = \int \frac{dx}{x}$$

$$v = \ln x + A$$

$$\frac{y}{x} = \ln x + A$$

$$y(x) = x\ln x + Ax$$

Ex2 Solve $(x-y)dy = (x+y)dx$

Soln

test for homogenity Let $y = vx$

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

$$\frac{d(vx)}{dx} = \frac{x+vx}{x-vx}$$

$$\frac{d(vx)}{dx} = \frac{1+v}{1-v}$$ (a function of v only)

∴ The equation is homogeneous

$$\frac{d(vx)}{dx} = \frac{1+v}{1-v}$$

$$v + x\frac{dv}{dx} = \frac{1+v}{1-v}$$

$$x\frac{dv}{dx} = \frac{1+v}{1-v} - \frac{v}{1}$$

$$x\frac{dv}{dx} = \frac{1+v^2}{1-v}$$

$$\frac{1-v}{1+v^2}\,dv = \frac{dx}{x}$$

$$\int\left(\frac{1}{1+v^2} - \frac{v}{1+v^2}\right)dv = \int\frac{1}{x}\,dx$$

$$\int\frac{dv}{1+v^2} - \int\frac{v}{1+v^2}\,dx = \ln x + A$$

$$\tan^{-1}(v) - \frac{1}{2}\ln(1+v^2) = \ln x + A$$

$$\tan^{-1}(y/x) - \frac{1}{2}\ln\left(1+\frac{y^2}{x^2}\right) = \ln x + A$$

$$\tan^{-1}(y/x) - \frac{1}{2}\ln\left(\frac{x^2+y^2}{x^2}\right) = \ln x + A$$

$$\tan^{-1}(y/x) - \frac{1}{2}\ln(x^2+y^2) = A$$

Exercise: Test for homogeneity and solve each equation

1. $x\,dy - (x+y)\,dx = 0$
2. $x\,dy + (2y-x)\,dx = 0$
3. $xyy' = x^2+y^2$
4. $xy' = y + \sqrt{x^2-y^2}$
5. $(y^2-xy)\,dx + (x^2+xy)\,dy = 0$

EXACT EQUATIONS

Consider the ODE

$$M\,dx + N\,dy = 0 \quad \cdots (1)$$

where M, N are respectively both functions of x and y. If there exists a function $u = u(x,y)$ which is a solution of the ODE (1) and further satisfies

$$u_x = M, \quad u_y = N$$

Then the ODE (1) is said to be exact. The function $u = u(x,y)$ is called the integral of the ODE (1).

Now, if the second partials of u are continuous, we obtain

$$\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial^2 u}{\partial y\,\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}\right)$$

$$\frac{\partial}{\partial y}(M) = \frac{\partial}{\partial x}(N)$$

or $$M_y = N_x \quad \cdots (2)$$

Thus, the ODE (1) is exact if (2) holds

Next, $u = u(x,y)$

$$\therefore \frac{du}{dx} = \frac{\partial u}{\partial y}\cdot\frac{dy}{dx} + \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial x}$$

$$= N\cdot\frac{dy}{dx} + M$$

$$du = M\,dx + N\,dy = 0$$

Ex1 Show that the equation $(2x+2y+1)dx+(2x-1)dy=0$ is exact and deduce its integral solution.

Soln

Compare the given equation with

$$Mdx+Ndy=0$$

we get

$M = 2x+2y+1$

$N = 2x-1$

Test for exactness

$M_y = 2$, $N_x = 2$

$\Rightarrow M_y = N_x$

Let $u = u(x,y)$ be the integral solution we obtain

$u_x = M$ ---(1)

$u_y = N$ ---(2)

from (1) we get

$u_x = 2x+2y+1$

$\int u_x dx = \int (2x+2y+1)dx$

$u = x^2+2xy+x+f(y)$ --(3)

Differentiating (3) w.r.t y

$u_y = 2x + f'(y)$ ---(4)

from (2) and (4) we get

$2x+f'(y) = 2x-1$

$\therefore f'(y) = -1$

$\Rightarrow f(y) = -y + \text{constant}$

Subst. in (3)

$\therefore u = x^2+2xy+x-y+\text{constant}$

OR

$u_x = M$ ---(1)

$u_y = N$ ---(2)

From (2) we get

$u_y = 2x-1$

$\int u_y = \int (2x-1)dy$

$u = 2xy - y + f(x)$ --(3)

Diff (3) w.r.t x

$u_x = 2y + f'(x)$ ---(4)

From (1) and (4)

$2y + f'(x) = 2x+2y+1$

$\Rightarrow f'(x) = 2x+1$

$f(x) = \int (2x+1)dx$

$f(x) = x^2+x+\text{constant}$

Subst. in (3)

$u = x^2+2xy+x-y+\text{constant}$

The required solution becomes

$x^2+2xy+x-y = \text{constant}$

Alternatively if the equation $Mdx+Ndy=0$ is exact its integral solution can be found by evaluting the integrals $\int M dx$ and $\int N dy$, and adding the two result but taking only one of similar terms (if any) in the two result.

Ex2: The ODE $(2x+2y+1)dx+(2x-1)dy=0$ has been shown in Ex 1 to be exact. Hence, its integral solution is given by

$\int(2x+2y+1)dx + \int(2x-1)dy = \text{constant}$

$\Rightarrow [x^2+2xy+x] + [2xy-y] = \text{constant}$

the solution becomes

$x^2+2xy+x-y = \text{constant}$

Exercise

Show that the equation is exact and find its solution

1) $(x-xy^2)dx + (2y-x^2y)dy = 0$
2) $y^2dx + 2xy\,dy = 0$
3) $(x^2-2xy+y)dx + (y^2-x^2+x)dy = 0$
4) $\frac{1}{y}dx + (1-x/y^2)dy = 0$
5) $(y\cos x + 2xe^{y^2})dx + (\sin x + x^2e^y + 2)dy = 0$

Ex3: Consider next the ODE

$ydx - xdy = 0$ --- (1)

test for exactness

$M = y$, $N = -x$

$M_y = 1$, $N_x = -1$

$M_y \neq N_x$

Therefore equation (1) is not exact.

Multiply (1) by $1/x^2$ to get

$y/x^2\,dx - \frac{1}{x}dy = 0$ --- (2)

for the ODE (2)

$M = y/x^2$, $N = -\frac{1}{x}$

$M_y = \frac{1}{x^2}$, $N_x = \frac{1}{x^2}$

$\Rightarrow M_y = N_x \Rightarrow$ Exactness

Thus eqtn (2) is exact

integrate eqtn (2) we get

$\int y/x^2\,dx + \int -\frac{dy}{x} = \text{constant}$

$-y/x - y/x = \text{constant}$

Take one of similar terms, the solution of eqtn (2) and consequently that of (1) becomes

$\frac{y}{x} = \text{constant}$

NOTE: (i) $\mu = \frac{1}{x^2}$ known as an integrating factor of the ODE (1)

ii) μ is not unique, it is verifable that

$\mu = \frac{1}{x^2}, \frac{1}{y^2}, \frac{1}{xy}, \frac{1}{x^2+y^2}$

can serve as μ for equation (1)

Generally, if the equation

$Mdx + Ndy = 0$ --- (1)

is not exact there may exist an integrating factor μ such as equation

$\mu Mdx + \mu Ndy = 0$

is exact.

ie $\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N)$ --- (2)

the condition (2) exactness

To get μ, the PDE (2) is more difficult than ODE (1).

In any cases the determination of an integrating factor is a matter of trial and error the three most

important method of investing an existing factor in simple cases are given below

1) If $\frac{M_y - N_x}{N}$ is a function of x only then there is an integrating factor μ which is a function of x only and is given by the ODE

$$\frac{d\mu}{dx} = \left(\frac{M_y - N_x}{N}\right)\mu$$

$$\Rightarrow \frac{d\mu}{\mu} = \left(\frac{M_y - N_x}{N}\right)dx$$

$$\Rightarrow \int \frac{d\mu}{\mu} = \int \frac{M_y - N_x}{N}\,dx$$

$$\ln \mu = \int \frac{M_y - N_x}{N}\,dx$$

$$\mu = e^{\int \left(\frac{M_y - N_x}{N}\right)dx}$$

Ex4: Consider again the ODE $ydx - xdy = 0$ ---① is inexact or not exact therefore, $M = y$, $N = -x$

$\Rightarrow M_y = 1 \neq N_x = -1$

and $\frac{M_y - N_x}{N} = \frac{-2}{x}$, is a function of x only

therefore there is an integral factor μ which is a function of x only and this μ is given by

$$\mu = e^{\int \frac{M_y - N_x}{N}dx}$$

$$\therefore \mu = e^{\int \frac{-2}{x}dx}$$

$$= e^{-2\ln x}$$

$$\mu = x^{-2} = \frac{1}{x^2}$$

Then multiply eqtn ① by $\mu = 1/x^2$ to get

$$\frac{y}{x^2}dx - \frac{dy}{x} = 0$$

Integrate to get the solution $\frac{y}{x} = \text{constant}$

2) If $\frac{N_x -$

μ is given

$\frac{d\mu}{dy} =$

$\Rightarrow \mu =$

Ex5: The equ

verify th

Therefore

a function

Then mul

$\frac{dx}{y}$

Integrate

3) If

3) If μ is

y only,

Solve (x

Verify

$\frac{M_y - N}{N}$

ie M

M_y

So take

$\mu = -1$

consequ

obtain

2 If $\frac{N_x - M_y}{M}$ then is a function of y only then μ is given by the ODE

$$\frac{d\mu}{dy} = \left(\frac{N_x - M_y}{M}\right)\mu$$

$$\Rightarrow \mu = e^{\int \left(\frac{N_x - M_y}{M}\right) dy}$$

Ex: The equation $y\,dx - x\,dy$ --- (1) is not exact verify that $\frac{N_x - M_y}{M} = \frac{-2}{y}$ is a function of y only

Therefore there is an integrating factor, μ which is a function of y only and this μ is given by

$$\mu = e^{\int \frac{N_x - M_y}{M} dy}$$

$$\mu = e^{\int \frac{-2}{y} dy} = e^{-2\ln y} = \frac{1}{y^2}$$

Then multiply eqtn (1) by $\mu = 1/y^2$ to get

$$\frac{dx}{y} - \frac{x}{y^2} dy = 0$$

Integrate to get the solution $x/y = \text{constant}$

3) If μ is neither a function of x only nor a function of y only, take $\mu = x^m y^n$ as a trial integrating factor

Ex: Solve $(x^2y - y)dx + x^2y^2 dy = 0$ --- (1)

soln

Verify that the ODE is not exact observe also that $\frac{M_y - N_x}{N}$ and $\frac{N_x - M_y}{M}$ are both function of x and y

ie $M = x^2y - y$ $\quad N = x^2y^2$

$M_y = x^2 - 1$ $\quad N_x = 2xy^2$

multiplying the ODE by $\mu = x^m y^n$ we obtain

$(x^{m+2}y^{n+1} - x^m y^{n+1})dx + (x^{m+2}y^{n+2})dy = 0$

So take $\mu = x^m y^n$ and that for exactness $m = -2$ and $n = -1$

consequently, $\mu = \frac{1}{x^2 y}$ with this integrating factor obtain the solution of (1) as $x - \frac{1}{x} + \frac{y^2}{2} = \text{constant}$.

4 Express a separable equation in the form of $P(x)\,q(y)\,dx + u(x)\,v(y)\,dy = 0$ and take $\mu = \frac{1}{u(x)\,q(y)}$ as a trial integrating factor

Ex 7: Solve $\frac{dy}{dx} = \frac{y^2-1}{x}$

Soln

$(y^2-1)\,dx - x\,dy = 0$

verify that the later equation is not exact.

Identify; $P(x) = 1$, $q(x) = y^2-1$, $u(x) = -x$, $v(y) =$

Hence, take $\mu = \frac{-1}{x(y^2-1)}$

multiply the ODE (2) by $\mu = \frac{-1}{x(y^2-1)}$ to get

$$-\frac{dx}{x} + \frac{dy}{y^2-1} = 0$$

So that, $M = \frac{-1}{x}$, $N = \frac{1}{y^2-1}$ $\Rightarrow M_y = 0 = N_x \Rightarrow$ exact

$\therefore -\ln x + \int \frac{dy}{y^2-1} = \text{constant}$

ie $-\ln x + \ln\sqrt{\frac{y-1}{y+1}} = \text{constant}$

5. Express a homogeneous 1st order ODE in the form $M\,dx + N\,dy$ and let $\mu = \frac{1}{xM+yN}$ be a trial integrating factor

Ex 8: Solve $\frac{dy}{dx} = \frac{x+y}{x-y}$

Soln

$(x+y)\,dx - (x-y)\,dy = 0$ - - - (*)

$M = x+y$, $N = -(x-y)$, $\Rightarrow M_y = 1 \neq N_x = -1$

$\Rightarrow$ the ODE Equation is not exact

take $\mu = \frac{1}{x(x+y) + y(-x+y)} = \frac{1}{xM+yN}$

$\mu = \frac{1}{x^2+xy-xy+y^2} = \frac{1}{x^2+y^2}$

multiplying eqtn (*) by $\mu = \frac{1}{x^2+y^2}$ to get

$$\left\{\frac{x+y}{x^2+y^2}\right\}dx - \left\{\frac{(x-y)}{x^2+y^2}\right\}dy = 0 \quad **$$

Verify that

$$M_y = N_x = \frac{x^2-y^2-2xy}{(x^2+y^2)^2}$$

and hence the ODE ** is exact.

Integrate equation (**) we get

$$\int\left(\frac{x}{x^2+y^2} + \frac{y}{x^2+y^2}\right)dx + \int\left(\frac{y}{x^2+y^2} - \frac{x}{x^2+y^2}\right)dy = \text{const.}$$

$$\left[\frac{1}{2}\ln(x^2+y^2) + \tan^{-1}\left(\frac{x}{y}\right)\right] + \left[\frac{1}{2}\ln(x^2+y^2) - \tan^{-1}\left(\frac{y}{x}\right)\right] = \text{constant}$$

take similar terms the solution becomes

$$\frac{1}{2}\ln(x^2+y^2) - \tan^{-1}\left(\frac{y}{x}\right) = 0$$

but $\tan^{-1}\left(\frac{x}{y}\right) = \frac{\pi}{2} - \tan^{-1}\left(\frac{y}{x}\right)$

taking similar terms the solution becomes

$$\frac{1}{2}\ln(x^2+y^2) - \tan^{-1}\left(\frac{y}{x}\right) + \frac{\pi}{2} = A$$

$$\frac{1}{2}\ln(x^2+y^2) - \tan^{-1}\left(\frac{y}{x}\right) = A - \frac{\pi}{2}$$

$$\frac{1}{2}\ln(x^2+y^2) - \tan^{-1}\left(\frac{y}{x}\right) = B$$

where $B = A - \frac{\pi}{2}$

6. If $M = yf(xy)$ and $N = xg(xy)$ then let $\mu = \frac{1}{xM - yN}$ as a trial integrating factor.

Ex 7. convert $y' = \frac{xy^2 - y}{x}$ ... (i) into an exact differential equation

soln

$$\frac{dy}{dx} = \frac{xy^2 - y}{x}$$

$$\Rightarrow x\,dy = (xy^2 - y)\,dx \quad \text{... (i)}$$

$$\Rightarrow (xy^2 - y)\,dx - x\,dy = 0$$

$$y(xy - 1)\,dx - x\,dy = 0$$

$M = xy^2 - y$, $N = -x$

$M_y = 2xy - 1 \neq N_x = -1$

Hence not exact

take $\mu = \frac{1}{x(xy^2 - y) - y(-x)} = \frac{1}{xM - yN}$

$$\mu = \frac{1}{x^2y^2 - xy + xy} = \frac{1}{x^2y^2}$$

multiplying eqn (i) ODE by $\mu = \frac{1}{x^2y^2}$ to get

$$\frac{xy - 1}{x^2y}\,dx - \frac{1}{xy}\,dy = 0 \quad \text{... (ii)}$$

verify that

$M_y = N_x = \frac{1}{x^2y^2}$ and hence the ODE (ii) is exact.

Exercise

Show that the equation is exact and find its solution

$\frac{-y^2}{x^2}dx + \frac{2y}{x}dy = 0$

i $(x^2 - 2xy)dx + (y^2 - x^2 + x)dy = 0$ ..

ii $\frac{dx}{y} + \left(1 - \frac{x}{y^2}\right)dy = 0$

2. Test for exactness in the equation and find an integrating factor if found inexact. Hence solve each equation

i $(1+xy)^2 dx + x^2 dy = 0$

ii $ydx - (x+y^2)dy = 0$

iii $(x^2 - y^2)dx + xydy = 0$

iv $xydx + (x^2+y)dy = 0$

v $(y^2+xy)dx - x^2 dy = 0$

3. Show that y/x is an integrating factor for the equation $2x \ln x \frac{dy}{dx} + y = 0$ and hence obtain its solution.

4. Show that the equation $(2xy + y^2)dx + (2x^2 + 3xy)dy = 0$ is inexact find an integrating factor and hence solve the equation.

5. Show that the equation is exact and find its solution

i $y^2 dx + 2xy dy = 0$

ii $(x - xy^2)dx + (2y - x^2 y)dy = 0$

6. Solve the equations

i $x^2 y^2 dx + (x + xy^2)dy = 0$

ii $(3xy + y^2)dx + (x^2 + xy)dy = 0$

iii $ydx + (2x - ye^y)dy = 0$

iv $dx + \left(\frac{x}{y} - \sin y\right)dy = 0$

v $x^2 y^3 dx + (x + xy^3)dy = 0$

LINEAR EQUATIONS

An ODE of the form

$$\frac{dy}{dx} + P(x)\,y = f(x)$$

where $P(x)$ and $f(x)$ are function of x only is said to be linear of the first order. This type of equation can be solved by the use of an integrating factor μ as follows

$$y' + Py = f(x) \quad \text{---} (1)$$

Multiply (1) by μ

$$\mu y' + \mu P y = \mu f(x) \quad \text{---} (2)$$

The function μ is chosen so that the L.H.S of (2) is an exact differential

ie L.H.S $= \frac{d}{dx}(\mu y)$

$$\therefore \mu y' + y\frac{d\mu}{dx} = \mu y' + \mu P y$$

$$\Rightarrow y\frac{d\mu}{dx} = \mu P y \quad \text{if } y \neq 0$$

$$\therefore \frac{d\mu}{\mu} = P\,dx$$

$$\int \frac{d\mu}{\mu} = \int P\,dx$$

$$\ln \mu = \int P\,dx$$

$$\mu = e^{\int P\,dx}$$

observe that μ is a function of x only.

Ex1 $y' - y = x$ solve the ODE.

Soln

comparing the ODE with eqtn (1)

ie $y' - y = x$

$y + Py = f(x)$

observe that $P = -1$

$$\therefore \mu = e^{\int P\,dx} = e^{\int -dx} = e^{-x}$$

multiple the ODE by $\mu = e^{-x}$

$$y'e^{-x} - ye^{-x} = xe^{-x}$$

$$\Rightarrow \frac{d}{dx}(ye^{-x}) = xe^{-x}$$

$$d(ye^{-x}) = xe^{-x}\,dx$$

$$ye^{-x} = \int xe^{-x}\,dx$$

$$= -xe^{-x} - \int -e^{-x}\,dx$$

$$ye^{-x} = -xe^{-x} - e^{-x} + A$$

$$\therefore y(x) = -x - 1 + Ae^{x}$$

Ex2 Solve the IVP $y' - 2y = e^{-x}$, $y(0) = 3$

Soln

$y' - 2y = e^{-x}$

$y' + Py = f(x)$

$$\mu = e^{\int P\,dx} = e^{\int -2\,dx} = e^{-2x}$$

multiplying the ODE by $\mu = e^{-2x}$

$$y'e^{-2x} - 2ye^{-2x} = e^{-(2x+x)}$$

$$y'e^{-2x} - 2ye^{-2x} = e^{-3x}$$

$$\frac{d}{dx}(ye^{-2x}) = e^{-3x}$$

$$ye^{-2x} = \int e^{-3x}\,dx$$

$$ye^{-2x} = -\frac{1}{3}e^{-3x} + A$$

$$y(x) = -\frac{1}{3}e^{-x} + Ae^{2x}$$

Now $y(0) = 3$

$$\Rightarrow \frac{1}{3}e^{-(0)} + Ae^{2(0)}$$

$$\Rightarrow -1/3 + A = 3 = 1$$

$\therefore y(x) = \frac{-1}{3}e^{-x} + \frac{10}{3}e^{2x}$

OR

$y(x) = \frac{1}{3}(10e^{2x} - e^{-x})$

Exercise

1. Solve the equation

i. $y' + y = e^{-x}$

ii. $y' - 5y = x$

iii. $y' - xy = 0$

iv. $xy' - y = x$

v. $y' + y = \sin x$

2. Solve the IVP

i. $xy' + 2y = x^2 - x + 1$, $y(1) = \frac{1}{2}$

ii. $xy' + y = e^x$, $y(1) = 1$

iii. $xy' + 2y = \sin x$, $y(\pi) = \frac{1}{\pi}$

iv. $y' - 2y = e^{2x}$, $y(0) = 2$

v. $y' + y = \frac{1}{1+x^2}$, $y(0) = 0$

If $p(x)$ is a constant then the equation $y' + py = f(x)$ is a constant coefficient linear first order ODE which has the general form

$y' + ay = f(x)$ ---- (1)

where "a" is a constant and $f(x)$ is a function of x only. Besides the general method of an integrating factor another method of solving a constant first order ODE is given below.

The homogenous equation associated with the inhomogenous eqtn (1) $y' + ay = f(x)$ is $y' + ay = 0$ ---- (2)

If y_c is the complementary soln of eqtn (2) and y_p is any particular soln of eqtn (1) it can be shown that $y(x) = y_c + y_p$

Since y_c contains one arbitrary constant, hence $y_c + y_p$ will also contain one arbitrary constant and hence will be the general solution of eqtn (1). Solution y_c is called the complementary function and y_p is a particular integral.

Let $y = e^{mx}$ be a trial soln of equation (2)

$\therefore y' = me^{mx}$ subst in eqtn (2)

ie $y' + ay = 0 \Rightarrow me^{mx} + ae^{mx} = 0$

$\Rightarrow e^{mx}(m + a) = 0$

Since $e^{mx} \neq 0$ hence $m + a = 0$

Thus, $y = e^{mx}$ is a solution of eqtn (2) if $m = -a$.

Thus solution is $y = e^{-ax}$

Consequently the general soln of eqtn (2) becomes $y_c = Ae^{-ax}$ where A = const

The eqtn (3) ie $m + a = 0$ is called an auxillary or characteristic equation of ODE (2)

Example 3: Write the general solution of the equation

i) $y' - y = 0$ (ii) $2y' + y = 0$

Soln

i) $y' - y = 0$
$y' + ay = 0$ } compare
observe that $a = -1$
from the auxillary equation
$m + a = 0$
$m - 1 = 0$
$m = 1$
$\therefore y_c = Ae^{-ax}$
$\Rightarrow y_c = Ae^{x}$

(ii) $2y' + y = 0$
$y' + \frac{1}{2}y = 0$
$y' + ay = 0$ } compare
$\therefore a = \frac{1}{2}$
From the auxillary equation
$2m + 1 = 0$
$2m = -1$
$m = -\frac{1}{2}$
$\therefore y_c = Ae^{-ax} = Ae^{-\frac{1}{2}x}$

Variation of parameter for the determination of a particular integral y_p for a given first order of an inhomogeneous ODE equation (1). The general solution of a homogeneous eqtn (2) associated with eqtn (1) is $y_c = Ae^{-ax}$

$y' + ay = f(x)$ - - - (1)
$y' + ay = 0$ - - - (2)

To get the particular integral y_p we substitute v for A in y_c so that $y_p = ve^{-ax}$ where v is a function of x

$\therefore y_p' = -ave^{-ax} + \frac{dv}{dx}e^{-ax}$

Subst. in the ODE eqtn (1) to get

$-ave^{-ax} + \frac{dv}{dx}e^{-ax} + ave^{-ax} = f(x)$

$\frac{dv}{dx}e^{-ax} = f(x)$

$dv = e^{ax} f(x)\,dx$

$v = \int e^{ax} \cdot f(x)\,dx$

$y_p = e^{-ax}\int e^{ax} \cdot f(x)\,dx$

Ex 4: Solve $y' - y = x$

Soln

AE: $m - 1 = 0$
$\Rightarrow m = 1$

CF: $y_c = Ae^x$

PI: $y_p = ve^x \Rightarrow y'$

$y'_p = ve^x + \frac{dv}{dx}e^x$

Subst. in the ODE

$ve^x + \frac{dv}{dx}e^x - ve^x = x$

$\frac{dv}{dx}e^x = x$

$dv = xe^{-x}dx$

$v = \int xe^{-x}dx$

$v = -xe^{-x} - e^{-x}$

$y_p = ve^x = (-xe^{-x} - e^{-x})e^x$

$y_p = -x - 1$

$y(x) = y_c + y_p$

$y(x) = Ae^x - 1 - x$

Ex: Solve $y' - 2y = e^{2x}$

Soln

AE: $m - 2 = 0$

$m = 2$

CF: $y_c = Ae^{mx} = Ae^{2x}$

PI: $y_p = ve^{2x}$

$y'_p = 2ve^{2x} + \frac{dv}{dx}e^{2x}$

Subst. in the ODE

$2ve^{2x} + \frac{dv}{dx}e^{2x} - 2ve^{2x} = e^{2x}$

$\frac{dv}{dx}e^{2x} = e^{2x}$

$\frac{dv}{dx} = 1$

$v = \int dx$

$v = x$

$y_p = ve^{2x} = xe^{2x}$

$y(x) = y_c + y_p$

$= Ae^{2x} + xe^{2x}$

Exercise

1. Use variation of Parameter to solve

i. $y' + y = e^{-x}$

ii. $y' + 3y = x + e^{-2x}$

iii. $y' + y = 1 + xe^{-x}$

iv. $\frac{dx}{dy} = e^y - x$

2. Solve, $\frac{dy}{dx} + 2y\tan x = \sin x$, $y(\pi/3) = 0$

3. Solve the following ODE's

i. $xy' - y = x^2$

ii. $y' + y\cot x = \cos x$

iii. $y' - y\tan x = \cos^2 x$

iv. $(1 + \sin x)y' + y\cos x = \tan x$

v. $(1 - x^2)y' - xy = 1$

4. A particle is projected upwards in a resisting medium and its velocity v after time t is given by $dv/dt + kv + g = 0$ where k, g are constants if the initial velocity is u. Show that the particle comes to rest after time $\frac{1}{k}\ln\left(1 + \frac{kv}{g}\right)$

Autonomous and non-Autonomous differential Equations

A differential equation in which the independent variable (x) does not appear explicitly is said to be an AUTONOMOUS DIFFERENTIAL EQUATION.

eg. $y' = 6 - y^3$, $y'' - y = 0$, $\frac{dy}{dx} = 5y + 9$, $y' = y + 1$

Thus in general, an autonomous first order is of the form $F(y, y') = 0$ or $\frac{dy}{dx} = h(y)$ (normal form).

Critical points of an Autonomous First order ODE

Recall in calculus, we discussed critical values and critical points of a function $y = f(x)$.

And we say that critical point will occur where $y' = 0$ or $\frac{dy}{dx} = 0$.

So for an Autonomous first order differential equation:

$$\frac{dy}{dx} = f(y) \quad - - - - (1)$$

its critical point will occur where $\frac{dy}{dx} = 0$ That is $f(y) = 0$ ie the zeros of f. we say that a real number c is a critical point of an autonomous D.E eqtn (1) if $f(c) = 0$.

We can also call a critical point, an equilibrium point, stationary point, Turning point.

NOTE: If c is a critical point of eqtn (1) then $y = c$ is a CONSTANT SOLUTION of the autonomous equation.

Singular Solution

Solution of a differential equation that cannot be obtained from the general solution gotten by the usual method of solving the differential equation. Thus, any particular solution that cannot be derived from the general solution is called a SINGULAR SOLUTION

1 For example; $(y')^2 = 4y$ has the general solution $y(x) = (x+c)^2$, the line $y=0$ is also a solution of the differential equation, but it is not a member of the family constituting the general solution.

2 The function $y = cx + \frac{1}{c}$ is also a one-parameter family of solutions for the differential equation $xy' + \frac{1}{y'} = y$. Any solution that is derived from this family is called a particular solution or integral

eg. $y = 1x + 1$ for $c = 1$
$y = 2x + \frac{1}{2}$ for $c = 2$

But $y^2 = 4x$ or $y = 2x^{1/2}$ is also a solution of the same equation.

To see this,

if $y^2 = 4x$ then $\frac{dy}{dx} = \frac{2}{y}$

$\therefore xy' + \frac{1}{y'} = \frac{2x}{y} + \frac{y}{2} = \frac{4x + y^2}{2y} = \frac{y^2 + y^2}{2y} = y$

OR if $y = 2x^{1/2}$, $y' = x^{-1/2}$

$\therefore xy' + \frac{1}{y'} = x \cdot x^{-1/2} + \frac{1}{x^{-1/2}} = x^{1/2} + x^{1/2} = 2x^{1/2} = y$

So $y^2 = 4x$ or $y = 2x^{1/2}$ is a particular solution.

Notice that no amount of value of c in $y = cx + \frac{1}{c}$ can ever yield $y^2 = 4x$ ~~thus~~

Thus $y^2 = 4x$ is not part of that one-parameter family $y = cx + \frac{1}{c}$. Hence $y^2 = 4x$ is called a singular solution of the ODE

A Tip on how to get a Singular Solution

Let consider the differential equation

$$\frac{dy}{dx} = g(x)h(y) \quad \text{---} \quad (1)$$

Thus $\frac{1}{h(y)}\frac{dy}{dx} = g(x)$

If y_0 is a root of $h(y) = 0$, then y_0 is a solution of equation (1) which may not be part of the general solution of eqtn (1)

Example 1: $\frac{dy}{dx} = 6x(y-1)^{2/3}$

observe that $h(y) = (y-1)^{2/3}$ and that $y = 1$ is a root of $h(y)$.

The general solution of this equation is $y = (x^2 + C)^3 + 1$

observe that $y = 1$ is also a solution that is not derivable from this general solution.

Thus $y = 1$ is a singular solution.

Application of First Order Linear Equation

1 Growth and Decay problems (Growth of an investment)

If the sum P Naira is invested for t years, with the amount A naira is accumulated at the end of the period when interest is compounded yearly at r% per annum is given by the compound interest formular

$A = P(1+r)^t$ if on the other hand if interest is ~~compounded~~ computed K times in a year the periodic rate becomes r/K% and the number of conversion period becomes Kt and so the amount accumulated in this case is

$$A = P(1 + r/K)^{Kt}$$

take log to base e

$$\ln A = \ln P + Kt \ln(1 + r/K)$$

differentiate w.r.t t

$$\frac{1}{A}\frac{dA}{dt} = K\ln(1+r/K)$$

$$\text{ie } \frac{1}{A}\frac{dA}{dt} = K\left[\frac{r}{K} - \frac{r^2}{K^2} + \cdots\right]$$

$$= r - \frac{r^2}{K^2} + \cdots \text{higher power of } r/K$$

as $K \to \infty$, we obtain

$\frac{dA}{dt} = rA$ thus the amount A accumulated by the principal P when interest is compounded continually at r% is given by $\frac{dA}{dt} = Ar$

Ex1 The sum of P is deposited in a bank which pays interest at the rate r% per annum compound continuously find

i) The time t required for the original sum to double in value

ii For what value of t is r = 5%

iii Find the interest rate that must decay if the intial investment is to double in 10 years

Soln

Since interest is anliannually compounded and the initially sum deposited is ₦P then we have to solve initial value problem

$$\frac{dA}{dt} = rA, \quad A(0) = P$$

$$\frac{dA}{A} = r\,dt$$

$$\ln A = rt + C$$

$$A = e^{rt} \cdot e^{C}$$

$$A = Ke^{rt}$$

2. Growth of Population:

A culture of yeast grows at a rate proportional to the amount present at, if doubles in 5hrs, how many times original amount will become in 20hrs. b) what time does it take to tripple.

Soln

Let Q be the quantity, and is the initial quantity

$$\frac{dQ}{dt} \propto Q$$

$$\Rightarrow \frac{dQ}{dt} = KQ \quad (K = \text{constant})$$

we have to solve IVP

$$\frac{dQ}{dt} = KQ$$

$$Q(0) = Q_0$$

ie $$\frac{dQ}{Q} = K\,dt$$

$$\ln Q = Kt + C$$

$$Q = e^{Kt} \cdot e^{C}$$

$$Q = Be^{Kt} \quad (B = e^{C})$$

$$Q/B = e^{Kt}$$

Now, $Q(0) = Q_0 \Rightarrow C = Q_0$

$$Q = Q_0 e^{Kt}$$

Now, $Q = 2Q_0$ when $t = 5$

$$2Q_0 = Q_0 e^{5K}$$

$$2 = e^{5K}$$

$$e^{K} = 2^{1/5}$$

$$Q = Q_0 (2^{1/5})$$

next, when $t = 20$

$$Q = Q_0 \cdot 2^4$$

$$Q = 16Q_0$$

= 16 times the original amount

When $Q = 3Q_0$

$3Q_0 = Q_0 2^{t/5}$

$\therefore 2^{t/5} = 3$

$\frac{t}{5}\ln 2 = \ln 3$

$t \ln 2 = 5 \ln 3$

$t = \frac{5 \ln 3}{\ln 2}$

3. Newton's Law of Cooling

It states that the temperature of a body changes at a rate which is proportional to the difference with which the temperature of a body and its surrounding medium.

NOTE: This law holds for difference of temperature 20°c or 36°F.

Let $\bar{T}$ = temperature of a body after time t

$\bar{T}_0$ < $\bar{T}$ is the temperature of the surrounding medium

$\frac{d\bar{T}}{dt} \propto (\bar{T} - \bar{T}_0)$

$\therefore \frac{d\bar{T}}{dt} = -K(\bar{T} - \bar{T}_0)$, since t is decreasing with time

$\frac{d\bar{T}}{\bar{T} - \bar{T}_0} = -K\,dt$

$\ln(\bar{T} - \bar{T}_0) = -Kt + A$

$\bar{T} - \bar{T}_0 = e^{-Kt} \cdot e^{A}$

$\bar{T} - \bar{T}_0 = Be^{-Kt}$

$\bar{T} = \bar{T}_0 + Be^{-Kt}$

Ex3 A thermometer which reads 80°F in a house is taken outside where the temperature of the air is 45°F. After 3 minutes the thermometer reading is 60°F Find

i) The reading of the thermometer 6 minutes after it was taken outside

ii) The time required for the temperature to drop from 80°F to 46°F

$$\bar{T} = \bar{T}_0 + Ae^{-kt}$$

initially, $\bar{T}_0 = 45$ when $t = 0$

$$\frac{d\bar{T}}{dt} = -k(\bar{T} - \bar{T}_0)$$

$$\bar{T}_0 = 80 \quad , \quad \bar{T}_3 = 60$$

but $\bar{T}_0 = 45$, implying the general solution is $\bar{T} = 45 + Ae^{-kt}$

Now $\bar{T}_0 = t_0 \Rightarrow 80 = 45 + Ae^{-k(t)}$

$$80 - 45 = Ae^{-k(0)}$$

$$A = 35$$

$$\bar{T} = 45 + 35e^{-kt}$$

$$\therefore 60 = 45 + 35e^{-3k}$$

$$15 = 35e^{-3k}$$

$$(3/7)^{1/3} = e^{-3k}$$

recall that $\bar{T} = 45 + 35(e^{-k})^t$

or $\bar{T} = 45 + 35(3/7)^{t/3}$

when $t = 6$

$$\bar{T} = 45 + 35(3/7)^{6/3}$$

$$\bar{T} = 45 + 35(3/7)^2$$

$$\bar{T} = 45 + 35(9/49)$$

$$\bar{T} = 45 + \frac{45}{7}$$

$$\bar{T} = 45 + 6.428$$

$$\bar{T} = 51.43^\circ F$$

ii Next, when $\bar{T} = 46$ we get

$$46 = 45 + 35(3/7)^{t/3}$$

$$1 = 35(3/7)^{t/3}$$

$$\frac{1}{35} = (3/7)^{t/3}$$

$$\ln(1/35) = t/3 \ln(3/7)$$

$$\ln 1 - \ln 35 = \frac{t}{3}[\ln 3 - \ln 7]$$

$$-3\ln 35 = t[\ln 3 - \ln 7]$$

$$-3\ln 35 = -t[\ln 7 - \ln 3]$$

$$t = \frac{3\ln 35}{\ln 7 - \ln 3}$$

Exercise -

1) The sum of 20,000 depos in a bank which pays interest of 5% per annum compounded continuously. Find

i) The time of investment is taken to double the s

ii) The value of investment after 20 years

iii) what interest rate would have tripple the original investment in 20 yea

2) The radio active substa composes at a rate proportio to the amount present. If the half life of the subst is 1500 years. How long w the substance take to re the quater of its original s

3) A rumour spreading in a community at rate proportiona to the number of people that heard it initially only 0.1% the population have heard rumour but the figure rose to 1½% after one day. Find

(i) what percentage of the community will hear the rumour after a further one day

(ii) After what time will half the community hear the rumour.

The Bernoulli's Equation

A non-linear first order ODE of form

$$\frac{dy}{dx} + Py = Qy^n \quad \text{---(1)} \quad (n>1)$$

where P & Q are function of x only is known as Bernoulli's equation.

The substitution $u = y^{1-n}$ reduces the equation to the linear form

$$\frac{1}{1-n} \cdot \frac{du}{dx} + Pu = Q$$

as follows,

$$u = y^{1-n} \quad \text{---(2)}$$

$$\therefore \frac{du}{dx} = \frac{d}{dx}(y^{1-n}) \cdot \frac{dy}{dx}$$

$$\frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx}$$

$$\therefore \frac{1}{1-n}\frac{du}{dx} = y^{-n}\frac{dy}{dx} \quad \text{---(3)}$$

Divide (1) by y^n to get

$$y^{-n}\frac{dy}{dx} + Py^{1-n} = Q \quad \text{---(4)}$$

Subst. (2) and (3) in (4)

$$\frac{1}{1-n} \cdot \frac{du}{dx} + Pu = Q$$

Ex1 Solve the equation $\frac{dy}{dx} + xy = xy^2$

Soln

Equation is Bernoulli compare with the standard Bernoulli equation we get

$$P = x, \quad Q = x, \quad n = 2$$

So let $u = y^{1-2} = y^{-1}$ ---(1)

$$\therefore \frac{du}{dx} = -y^{-2}\frac{dy}{dx} \quad \text{----(2)}$$

Divide the ODE by y^2 to get

$$y^{-2}\frac{dy}{dx} + xy^{-1} = x \quad \text{---(3)}$$

use (1) & (2) in (3)

$$-\frac{du}{dx} + xu = x$$

$$\frac{du}{dx} - xu = -x$$

$$u' - xu = -x \quad \text{---} \quad (4)$$

$$\mu = e^{\int p\,dx} = e^{\int -x\,dx} = e^{-x^2/2}$$

multiplying (4) by $\mu = e^{-x^2/2}$

$$u'e^{-x^2/2} - xue^{-x^2/2} = -xe^{-x^2/2}$$

$$\frac{d}{dx}(ue^{-x^2/2}) = -xe^{-x^2/2}$$

$$ue^{-x^2/2} = \int -xe^{-x^2/2}\,dx$$

$$ue^{-x^2/2} = e^{-x^2/2} + A$$

$$u = 1 + Ae^{x^2/2}$$

$$\frac{1}{y} = 1 + Ae^{x^2/2}$$

Observe that the Bernoulli equation of Ex 1 is separable

$$\frac{dy}{dx} + xy = xy^2$$

$$\frac{dy}{dx} = x(y^2 - y)$$

$$\frac{dy}{y^2 - y} = x\,dx$$

$$\frac{dy}{y(y-1)} = x\,dx$$

$$\int\left(\frac{1}{y-1} - \frac{1}{y}\right)dy = \int x\,dx$$

$$\ln\left(\frac{y-1}{y}\right) = \frac{x^2}{2} + A$$

$$\frac{y-1}{y} = Be^{x^2/2} \quad (B = e^A)$$

$$1 - \frac{1}{y} = Be^{x^2/2}$$

$$\frac{1}{y} = 1 + Ce^{x^2/2} \quad (C = -B)$$

SECOND

In linear
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SECOND ORDER LINEAR ODE WITH CONSTANT COEFFICIENTS

An linear first order ODE as discussed earlier is a special case of the general linear ODE of the order n is of the form $a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_{n-1} y' + a_n y = f(x)$ where $a_0, a_1, a_2, \cdots, a_n$ and $f(x)$ are given functions of x or are constants. ODE is said to be homogeneous if $f(x) = 0$ and to be inhomogeneous if $f(x) \neq 0$ we are at this stage interested in the second order ODE with constant coefficient which have the form $ay'' + by' + cy = f(x)$ - - - (1) by constant coefficient we mean the a, b, c are not variables

The homogeneous or reduced equation corresponding to eqtn (1) is $ay'' + by' + cy = 0$ - - - (2)

If $y = e^{mx}$ be a trial solution of eqtn (2) then $y' = me^{mx}$, $y'' = m^2 e^{mx}$ consequently we obtain by substituting for y'', y' and y in equation (2) to get

$(am^2 + bm + c) e^{mx} = 0$, since $e^{mx} \neq 0$ implies

$am^2 + bm + c = 0$ - - - (3)

$\therefore y = e^{mx}$ is a solution of eqtn (2) if eqtn (3) is satisfied. Equation (3) is called the auxillary or the characteric equation of eqtn (2) and eqtn (3) is a quadratic equation in m and so has 2 roots, real or imaginary. Thus on each characteristic root of the characteristic eqtn of the ODE eqtn (2) there is a corresponding solution.

So let y_1 and y_2 be a solution of equation (2). Solutions are said to be linearly independent if whenever $c_1 y_1 + c_2 y_2 = 0$ where c_1 & c_2 are arbitrary constant, then $c_1 - c_2 = 0$ $\therefore c_1 = c_2$

Let y_1 and y_2 be two linear independent solution of equation (2). The linear combination $y = c_1 y_1 + c_2 y_2$, where c_1 and c_2 are arbitrary constant. This is also the solution of eqtn (2).

Generally $y_1, y_2, \cdots, y_n$ are n linearly independent solution of an nth order linear ODE the linear combination

$y = c_1y_1 + c_2y_2 + c_3y_3 + \cdots + c_ny_n$, where $c_1, c_2, \cdots, c_n$ are arbitrary constant is also a solution of the nth order linear ODE. The general solution of an nth order linear ODE must contain n arbitrary constant.

It is reasonable to take $y = c_1y_1 + c_2y_2$ as the general solution of eqtn (2) where y_1 and y_2 are linearly independent solution.

Let $M = m_1, m_2$ be the 2 roots of equation (3) there are 3 cases.

CASE I: The roots of eqtn (3) are real and different then $y_1 = e^{m_1x}$, $y_2 = e^{m_2x}$ and the general solution becomes $y = Ae^{m_1x} + Be^{m_2x}$ where A, B are arbitrary constants

Example 1: Solve $y'' - y' - 2y = 0$

Soln

AE: $y'' - y' - 2y = 0$

AE: $m^2 - m - 2 = 0$

$(m+1)(m-2) = 0$

$m = -1$ or $m = 2$

$y_1 = e^{-x}$ $y_2 = e^{2x}$

∴ General solution becomes

$y = Ae^{-x} + Be^{2x}$

CASE II: The roots of eqtn (3) are real and equal.

Let $M = m_1 = m_2$ the general solution becomes

$y = Ae^{mx} + Be^{mx}$

$= (A+B)e^{mx}$

$y = ce^{mx}$ $(c = A+B)$

This does not qualify for a general solution since it has only one arbitrary constant name $c = A+B$.

Another linearly independent solution is solved by writting $y = ue^{mx}$

$y' = mue^{mx} + u'e^{mx}$

$y'' = u''e^{mx} + 2mu'e^{mx} + um^2e^{mx}$

by subst. in eqtn (2) we obtain

$$ay'' + by' + cy = 0 \quad \text{---} \quad (2)$$

$$a(u'' + 2mu' + um^2)e^{mx} + b(mu + u')e^{mx} + cue^{mx}$$

$$au'' + (2am + b)u' + (am^2 + bm + c)u = 0 \quad \text{---} \quad (4)$$

but $am^2 + bm + c = 0$ from eqtn (3) further more the sum of roots of eqtn (3) is

$$2m = -\frac{b}{a} \Rightarrow 2am + b = 0$$

Thus, eqtn (4) becomes

$$y'' = 0 \text{ or } \frac{d^2u}{dx^2} = 0 \Rightarrow u = Ax + c$$

The general equation now becomes

$$y = y_1 + y_2$$

$$y = e^{mx} + (Ax + c)e^{mx}$$

$$y = (1 + Ax + c)e^{mx}$$

$$y = (Ax + B)e^{mx} \quad (\text{where } B = c + 1)$$

Example 2: Solve $4y'' - 4y' + y = 0$

Soln

$$4y'' - 4y' + y = 0$$

AE: $4m^2 - 4m + 1 = 0$

$$(2m - 1)^2 = 0$$

$$2m = 1$$

$m = \frac{1}{2}$ twice

The general solution becomes

$$y = (Ax + B)e^{x/2}$$

CASE III: Roots of eqtn (3) are imaginary

Let the roots be $m_1 = p + iq$, $m_2 = p - iq$

The general solution becomes

$$y = Ae^{(p+iq)x} + Be^{(p-iq)x}$$

$$y = e^{px}\{Ae^{iqx} + Be^{-iqx}\} \quad (e^{i\theta} = \cos\theta + i\sin\theta)$$

$$y = e^{px}\{A\cos qx + Ai\sin qx + B\cos qx - Bi\sin qx\}$$

$y = e^{px}[(A+B)\cos qx + i(A-B)\sin qx]$

$y = e^{px}[E\cos qx + F\sin qx\}$ where $A+B=E$, $F=A-B$

1 **Example 3:** Solve (i) $y''+y=0$ (ii) $y''+y'+y=0$

Soln

i AE: $y''+y=0$

$m^2+1=0$

$m^2=-1$

$m=\pm\sqrt{-1}$

$m=\pm i$

General solution is

$y = e^{0x}(A\cos x + B\sin x)$

$y = A\cos x + B\sin x$

(ii) $y''+y'+y=0$

$m^2+m+1=0$

$m=\frac{-1\pm\sqrt{1-4}}{2}$

$=-\frac{1}{2}\pm\sqrt{3}/2\,i$

General solution is

$y = e^{-1/2x}\{A\cos\frac{\sqrt{3}}{2}x + B\sin\frac{\sqrt{3}}{2}x$

Exercise: Solve

i $y''-y=0$

ii $y''-y'=0$

iii $y''-3y'+2y=0$

iv $y''-5y'+6y=0$

v $y''+w^2y=0$

(vi) $y''+2y'+2y=0$

(vii) $y''+2y'+4y=0$

(viii) $4y''+4y'+y=0$

(ix) $3y''-4y'+y=0$

(x) $y''-(a+b)y'+aby=0$

Generisation

The nth order linear homogenous ODE

$a_0y^{(n)} + a_1y^{(n-1)} + \cdots + a_{n-1}y' + a_ny = 0$ ---(5)

where $a_0 \neq 0$, this n^{th} order equation has auxillary or charaterestic equation

$a_0m^n + a_1m^{n-1} + \cdots + a_{n-1}m + a_n = 0$ ---(6)

Equation (6) solution as said before is $y = c_1y_1 + c_2y_2 + \cdots$ where $y_1, y_2, \cdots, y_n$ are the linearly independent solut corresponding to the n roots of its charaterestic equat (6) and $c_1, c_2, c_3, \cdots, c_n$ are arbitrary constant

The examples will now be given to illustrate the general approach of writting the general solution of a ODE or finding an ODE that has the general solutio

Example 4: Find the general solution of the linear homogenous ODE which has characteristic roots are 1, 2, -1, 3

Soln

The linearly independent solution are
$y_1 = e^x$, $y_2 = e^{2x}$, $y_3 = e^{-x}$, $y_4 = e^{3x}$ the general s/n becomes $y = Ae^x + Be^{2x} + Ce^{-x} + De^{3x}$

Example 5: Some roots are repeated; find the general solution of the linear homogenous ODE which has the characteristic roots 1, 1, 1, -2, -2, 2

Soln

$y_1 = e^x$, $y_2 = xe^x$, $y_3 = x^2e^x$, $y_4 = e^{-2x}$, $y_5 = xe^{-2x}$, $y_6 = e^{2x}$
the general solution becomes
$y = (A + Bx + Cx^2)e^x + (D + Ex)e^{-2x} + Fe^{2x}$

Example 6: Some roots are complex: find the general solution of the linear homogenous ODE which has the characteristic roots $1, 2, 1 \pm 2i, -2 \pm i\sqrt{3}$

Soln

$y_1 = e^x$, $y_2 = e^{2x}$, $y_3 = e^x \cos 2x$, $y_4 = e^x \sin 2x$, $y_5 = e^{-2x} \cos \sqrt{3}x$
$y_6 = e^{-2x} \sin \sqrt{3}x$ — the general solution becomes
$y = Ae^x + Be^{2x} + e^x(C\cos 2x + D\sin 2x) + e^{-2x}(E\cos \sqrt{3}x + F\sin \sqrt{3}x)$

Example 7: Find the linear homogenous equation whose characteristic roots are -1 and 2

Soln

Auxillary roots are $m = -1$ and $m = 2$
$\Rightarrow (m+1) = 0$ and $(m-2) = 0$
$\Rightarrow (m+1)(m-2) = 0$
$\Rightarrow m^2 - m - 2 = 0$
$y'' - y' - 2y = 0$

Example 8: Find the ODE which has the general solution. $y(x) = Ae^x + B\cos x + C\sin x$

Soln

$y_1 = e^x$, $y_2 = \cos x$, $y_3 = \sin x$

$m_1 = 1$, $m_2 = i$, $m_3 = -i$

$m - 1 = 0$ and $m = \pm i$

$m = \pm\sqrt{-1}$

$m^2 = -1$

$m^2 + 1 = 0$

$(m-1)(m^2+1) = 0$

$m^3 + m - m^2 - 1 = 0$

$m^3 - m^2 + m - 1 = 0$

$y''' - y'' + y' - y = 0$

1. Exercise: Solve

i. $y'' - 2y' - 3y = 0$

ii. $y''' - 2y'' + y' - 2y = 0$

iii. $y''' - 5y'' - 5y' - 3y = 0$

2. Find the ODE which has characteristic roots

i. $1, -2, -3, -3$ (iv) $3, 1 \pm \sqrt{2}\,i$

ii. $-1, -1, -1, 2, 2$ (v) $1, 1, -1, \frac{1}{2} \pm i\frac{\sqrt{3}}{2}$

iii. $-1, -1, \pm 2i$

3. Find the ODE which has the general solution

i) $y = A + Be^{-x}$ (iv.) $y = A + B\cos\sqrt{2}x + C\sin\sqrt{2}x$

ii. $y = (A + Bx)e^{-x}$ (v) ~~$y = e^x(A+Bx) + C\cos x + D\sin x +$~~

iii $y = Ae^{-2x} + e^x(B\cos 2x + C\sin 2x)$

v. $y = e^x(A + Bx) + C\cos x + D\sin x + E\cos\sqrt{3}x + F\sin\sqrt{3}x$

4. Solve the IVP

i) $y'' + 5y' + 6y = 0$
$y(0) = 1$
$y'(0) = 0$

(ii) $y''' - 2y'' + y' - 2y = 0$
$y(0) = 1$
$y'(0) = 0$
$y''(0) = 0$

Test for linear independence of solution

The solution $y_1, y_2, \cdots, y_n$ of an nth order linear ODE are linearly independent if $c_1y_1 + c_2y_2 + \cdots + c_ny_n = 0$

$\Rightarrow c_1 = c_2 = \cdots = c_n = 0$. So let,

$$c_1y_1 + c_2y_2 + \cdots + c_ny_n = 0$$

$$\Rightarrow c_1y_1' + c_2y_2' + \cdots + c_ny_n' = 0$$

$$c_1y_1'' + c_2y_2'' + \cdots + c_ny_n'' = 0$$

$$\vdots$$

$$c_1y_1^{(n-1)} + c_2y_2^{(n-1)} + \cdots + c_ny_n^{(n-1)} = 0$$

For trivial solution of the n-equation the coefficient

$$W = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & & & \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix} \neq 0$$

This determinant which is usually denoted by $W(x)$ is called (known as) the Wronskian of the nth order ODE. So the solution of an nth order ODE are linearly independent if the $W(x) \neq 0$

Example 9: Show that the function $y_1 = e^x$, $y_2 = e^{2x}$ are linearly independent

Soln

Function are linearly independent if

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} \neq 0 \Rightarrow \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} \Rightarrow 2e^{3x} - e^{3x} = e^{3x}$$

$\therefore e^{3x} \neq 0$; hence the functions are linearly independent

Exercise:

1. Verify that the solution of the ODE are linearly independent

i) $y'' - 3y' + 2y = 0$ (ii) $y'' + 2y + y = 0$ (iii) $y'' + 4y = 0$

w $y''+2y'+2y=0$ (v) $y'''+2y''=0$ (vi) $2y'''-y''+4y'-4y=0$

2. Show that the function are linearly independent

i $y_1=e^x$, $y_2=e^{2x}$, $y_3=e^{3x}$

ii $y_1=e^x$, $y_2=e^{-x}$, $y_3=\cosh x$

Second Order Linear Inhomogenous Equation with Constant Coefficients

The general form is

$$ay''+by'+cy=f(x) \quad ---(1)$$

where a, b, c are constants and $f(x)\neq 0$ is a function of x.

The general solution of this equation is made up of two parts one being the general solution y_c of the reduced homogenous equation

$ay''+by'+c=0$ ---(2) associated with it, and the order y_p been any particular solution which can be find by inspection or otherwise it can been shown that the linear combination $y=y_c+y_p$ ---(3) is a solution of the inhomogenous eqtn (1).

Now since y_c contains two arbitrary constant $y=y_c+y_p$ will also contain two arbitrary constant and hence will be the general solution of the inhomogenous eqtn (1).

The solution $y=y_c$ of eqtn (2) is called complementary function and the particular solution $y=y_p$ of eqtn (1) is called the particular integral of the inhomogenous eqtn.

Methods of finding the complementary function as already been covered. So we now jump to obtaining particular integral in certain specified cases

THE METHOD OF INDETERMINED COEFFICIENT

1. If $f(x)$ = a constant, then assume P.I of form $y_p=C$ (a constant)

Example 10: Find the PI of the equation $y'' - y' - 2y = 12$

Soln

Let $y_p = C$ (a constant) $\Rightarrow y_p' = y_p'' = 0$

Subst. in the ODE we get

$-2y_p = 12 \Rightarrow y_p = -6$

but $y_c = Ae^{-x} + Be^{2x}$

The general solution becomes $y = y_c + y_p$

$y(x) = Ae^{-x} + Be^{2x} - 6$

2. If $f(x)$ is a polynomial of degree n, and if the auxiliary equation has $m = 0$ as a root of multiplicity K, then assume a particular integral of form

$y_p = x^K (a_0x^n + a_1x^{n-1} + \dots + a_n)$ and by substitution the inhomogeneous equation varies on the constant $a_0, a_1, a_2, \dots, a_n$ are determined by equating coefficient.

If $m = 0$ is not a root of the auxiliary equation then $K = 0$

Example 11: Solve $y'' + 4y = x^2 + 2x + 3$

Soln

AE: $m^2 + 4 = 0$

$m^2 = -4$

$m = \pm\sqrt{-4}$

$= \pm 2i$

CF: $y_c = A\cos 2x + B\sin 2x$

PI: The value $m = 0$ is not a root of the AE implies $K = 0$

so we assume $y_p = x^0(a_0x^2 + a_1x + a_2)$ we obtain

$y' = 2a_0x + a_1$, $y'' = 2a_0$

Subst. in the ODE we get

$2a_0 + 4(a_0x^2 + a_1x + a_2) = x^2 + 2x + 3$

equate coefficient

$2a_0 + 4a_0x^2 + 4a_1x + 4a_2 = x^2 + 2x + 3$

$4a_0x^2 = x^2$
$4a_1x = 2x$
$2a_0 + 4a_2 = 3$

$\Rightarrow$

$4a_0 = 1$
$4a_1 = 2$
$2a_0 + 4a_2 = 3$

$\Rightarrow$

$a_0 = 1/4$
$a_1 = 1/2$
$a_2 = 5/8$

The general solution; $y = y_c + y_p$

$y = A\cos 2x + B\sin 2x + \frac{1}{4}x^2 + \frac{1}{2}x + 5/8$

Example 12: Solve $y'' - y' = x - 2$

Solution

AE: $m^2 - m = 0 \Rightarrow m(m-1) = 0$

$m = 0$ or $m - 1 = 0$

$m = 1$

CF: $y_c = A + Be^x$

PI: $m = 0$ is a root of the AE is of multiplicity 1

So. let, $y_p = x(a_0x + a_1) = a_0x^2 + a_1x$

$y_p' = 2a_0x + a_1$, $y_p'' = 2a_0$

Subst. in the ODE

$2a_0 - (2a_0x + a_1) = x - 2$

$2a_0 - a_1 = -2$ $\Rightarrow$ $2a_0 + a_1 = 2$ $\Rightarrow$ $a_0 = -1/2$

$-2a_0x = x$ $\Rightarrow$ $-2a_0 = 1$ $\Rightarrow$ $a_1 = 1$

$\therefore y_p = -\frac{1}{2}x^2 + x$

General solution becomes: $y = y_c + y_p$

$y = A + Be^x - \frac{1}{2}x^2 + x$

3. If $f(x) = ae^{bx}$, where a, b are constants and if the auxiliary equation has $m = b$ as a root of multiplicity K, then assume

$y_p = \lambda x^K e^{bx}$

where λ is a constant if $m = b$ is not a root, then $K = 0$

Example 13: Solve $\frac{d^2x}{dt^2} - \frac{dx}{dt} - 2x = e^{2t}$

Solution

AE: $m^2 - m - 2 = 0$

$m^2 + m - 2m - 2 = 0$

$m(m+1) - 2(m+1) = 0$

$(m-2)(m+1) = 0$

$m = -1$ or $m = 2$

CF: $x_c = Ae^{-t} + Be^{2t}$

PI: The value $m = 2$ is a root of the auxillary equation of multiplicity $k = 1$

So let $x_p = \lambda t e^{2t}$

$\therefore x_p' = 2\lambda t e^{2t} + \lambda e^{2t}$

$x_p'' = 4\lambda t e^{2t} + 2\lambda e^{2t} + 2\lambda e^{2t}$

Subst. in the ODE

$4\lambda t e^{2t} + 4\lambda e^{2t} - 2\lambda t e^{2t} - \lambda e^{2t} - 2\lambda t e^{2t} = e^{2t}$

$3\lambda e^{2t} = e^{2t}$

$3\lambda = 1$

$\lambda = 1/3$

$\therefore x_p = \frac{1}{3} t e^{2t}$

General solution becomes $x = x_c + x_p$

$x(t) = Ae^{-t} + Be^{2t} + \frac{1}{3} t e^{2t}$

4. If $f(x) = a\cos bx$ or $a\sin bx$, and if $m^2 + b^2$ is a factor of the auxiliary equation of multiplicity k, assumed

$y = x^k(\lambda \cos bx + \mu \sin bx)$

If $m^2 + b^2$ is not a factor, then $k = 0$

Example 14: Solve $y'' - 3y' + 2y = \cos x$

Soln

AE: $m^2 - 3m + 2 = 0$

$m^2 - m - 2m + 2 = 0$

$m(m-1) - 2(m-1) = 0$

$(m-2)(m-1) = 0$

$m = 2$ or $m = 1$

CF: $y_c = Ae^x + Be^{2x}$

PI: $m^2 + 1$ is not a factor of the AE implying that $k = 0$

$\therefore y_p = x^0(\lambda \cos x + \mu \sin x) = \lambda \cos x + \mu \sin x$

$y_p' = -\lambda \sin x + \mu \cos x$, $y_p'' = -\lambda \cos x - \mu \sin x$

Subst. in the ODE to get

$-\lambda \cos x - \mu \sin x - 3(-\lambda \sin x + \mu \cos x) + 2(\lambda \cos x + \mu \sin x) = \cos x$

Equating coefficient $-\lambda \cos x - \mu \sin x + 3\lambda \sin x - 3\mu \cos x + 2\lambda \cos x + 2\mu \sin x = \cos x$

$\lambda \cos x + \mu \sin x + 3\lambda \sin x - 3\mu \cos x = \cos x$

$(\lambda - 3\mu)\cos x + (\mu + 3\lambda)\sin x = \cos x$

Equating coefficients

$\lambda - 3\mu = 1 \Rightarrow \lambda = 1 + 3\mu$

$\mu + 3\lambda = 0$ subst. $\Rightarrow \mu + 3(1+3\mu) = 0$

$\mu + 3 + 9\mu = 0$

$10\mu = -3$

$\mu = -3/10$

$\lambda = 1 - \frac{9}{10}$

$\lambda = \frac{1}{10}$

$y_p = \frac{1}{10}\cos x - \frac{3}{10}\sin x$

The General solution becomes

$y(x) = y_c + y_p$

$y(x) = Ae^x + Be^{2x} + \frac{1}{10}\cos x - \frac{3}{10}\sin x$

1. Method of Undetermined coefficient
2. Method of variation of parameters

Example: Solve $y' - y = x$ method of undetermined coefficient

AE: $m - 1 = 0$

$m = 1$

CF: $y_c = Ae^x$

PI: let $y_p = a_0 x + a_1$

$y' = a_0$

Subst. in the DE

$a_0 - a_0 x - a_1 = x$

equating coefficient

$a_0 - a_1 = 0 \Rightarrow a_0 = a_1$

$-a_0 = 1 \Rightarrow a_0 = -1 = a_1$

$y_p = -x - 1$

$y = y_c + y_p$

$y(x) = Ae^x - x - 1$

ii Variation of constants (parameter)

$y' + y = x$

soln

$y_c = Ae^x$

Let $y_p = ve^x$, $y_p' = ve^x + e^x \frac{dv}{dx}$

Subst in the ODE

$ve^x + e^x \frac{dv}{dx} - ve^x = x$

$e^x \frac{dv}{dx} = x$

$\frac{dv}{dx} = xe^{-x}$

$\int dv = \int xe^{-x} dx$

$v = -xe^{-x} - e^{-x}$

$y_p = ve^x = e^x(-xe^{-x} - e^{-x}) = -x - 1$

$y(x) = y_c + y_p$

$= Ae^x - x - 1$

THE D - Operator

In the study of ODE constant coefficients the symbol "D" is often used to represent the differential operator $D = \frac{d}{dx}$

Hence, $Dy = y'$, $D^2y = y''$, $D^3y = y'''$ ... in terms of the D operator the second order constant coefficient inhomogenous ODE have the general form

$aD^2y + bDy + cy = f(x)$ OR $(aD^2 + bD + c)y = f(x)$

OR simply

$F(D)y = f(x)$ where $F(D) = aD^2 + bD + c$ is a polynomial operator in D of degree 2.

D operator satisfies the rules of differentiation hence can be treated like ordinary algebraic quantity despite the fact that it is an operator which has no numerical values.

Since D behaves like an algebraic quantity so must $F(D)$.